

Manuscript structure and Methods

Working title. Regulatory architecture of the feline renal cortex and the tissue-specificity shift underlying the chronic kidney disease transcriptome

Target. BMC Genomics or Scientific Reports. Fallback: Veterinary Sciences, Frontiers in Veterinary Science.

Framing. Exploratory, hypothesis-generating secondary analysis and resource description. No prespecified analysis plan; analytical decisions responsive to the data are disclosed in Methods.

1. Proposed structure

Introduction (4 paragraphs). Feline CKD as the leading cause of mortality in aged cats and the existing transcriptomic description of metabolic reprogramming. The gap: the cat now has a genome-wide map of regulatory regions across seven tissues, produced to interpret non-coding variants, but it has never been used to interpret disease expression. What bulk tissue can and cannot resolve, and why the previous reanalysis of the same dataset concluded that cell composition may contribute materially without measuring it. Objectives: annotate the cis-regulatory architecture of genes expressed in the feline renal cortex, test whether the CKD expression signature has recoverable coregulatory structure, and characterise the tissue breadth of the regulatory elements associated with genes that change.

Results (five subsections, matching Figures 1–4).

1. Cis-regulatory annotation of the feline renal cortex, including the assembly discordance and its detection.
2. Differential expression and reproduction of the source study.
3. Coexpression structure is not reproducible; the single stable axis tracks tubular content.
4. Motif enrichment is largely null; the ERR observation and the controls that reduce it to constitutive promoter architecture, except for a residual association with repression.
5. Repressed genes are enriched for kidney-specific enhancers and induced genes for ubiquitous ones.

Discussion (7–8 paragraphs). Principal finding stated as a shift in regulatory breadth. Consistency with cell composition change and why the two are not separable in bulk tissue. Relation to the two prior analyses of the same dataset. The ERR observation as constrained. Value of the annotation as a resource. The assembly caveat as a practical warning. Limitations. Generalisability.

Conclusions. Short, thematic, on what bulk renal transcriptomes in the cat can support and what requires cell-resolved data.

2. Methods

2.1 Data sources

Gene expression data were obtained from the multi-omics study of naturally occurring feline CKD deposited under GEO accession GSE303653 and published by Li and colleagues (doi:10.1038/s42003-025-09164-8). Rather than reprocessing the raw sequencing data, we used the normalised gene-level expression matrices distributed as

Supplementary Data 3 (renal cortex) and Supplementary Data 4 (renal medulla) of that publication. In the source study, total RNA was extracted from renal cortex and medulla of cats classified by IRIS stage, libraries were prepared with an Illumina stranded mRNA protocol and sequenced on a NextSeq 2000 platform to approximately 100 million reads per sample, reads were aligned to *Felis_catus_9.0* with RNAstar v2.5.3 in two-pass mode, per-gene counts were generated with htseq-count v2.16.2, counts were converted to counts per million with edgeR, genes with fewer than 2 counts per million in at least 5 samples were discarded, and the remaining values were normalised by the trimmed mean of M-values method. The deposited matrices are therefore log2-transformed, TMM-normalised counts per million indexed by gene symbol, not raw counts. The cortex matrix contained 15,149 genes across 21 samples (6 control, 8 CKD 1/2, 7 CKD 3/4) and the medulla matrix 15,267 genes across 18 samples (6 control, 7 CKD 1/2, 5 CKD 3/4). Sample identifiers, disease group, sex, age and breed are embedded in the header rows of each sheet. Group means computed from the age row reproduce the values reported in the demographic table of the source publication to one decimal place in every group and compartment, confirming that the metadata rows correspond to the published cohort. The deposited cortical matrix contains 21 of the 22 cortical samples present in the GEO series; the remaining sample was excluded by the source authors. Cohort characteristics are given in Table 1.

Chromatin data were obtained from GEO accession GSE182952 (BioProject PRJNA758414), which reports genome-wide mapping of regulatory regions in seven tissues of adult male domestic shorthair cats with two biological replicates per tissue. The series is deposited without an associated publication, so the processing details below are taken from the sample metadata of the deposit rather than from a peer-reviewed methods section. Sample metadata identify the renal samples as kidney cortex. We used the final replicated peak sets distributed as series-level supplementary files for CTCF, H3K27ac and H3K4me3 in all seven tissues (heart, kidney, liver, lung, small intestine, testis, thyroid). In the source study, chromatin immunoprecipitation followed the ENCODE3 pipeline v1 specification for histone modifications, reads were mapped with Bowtie2 v2.4.2, relaxed peaks were called with MACS2 v2.2.5, and peaks from the pooled replicate set were retained in the final replicated set when they overlapped by at least half their length with peaks from both biological replicates. Peak files are not filtered against a blacklist because none exists for *Felis catus*.

Genome annotation and sequence were obtained from the NCBI RefSeq assembly GCF_018350175.1 (*F.catus_Fca126_mat1.0*), NCBI *Felis catus* Annotation Release 105, comprising 39,307 genes of which 20,453 are protein-coding. The corresponding assembly report and genomic FASTA were downloaded from the same accession.

2.2 Assembly discordance and coordinate harmonisation

The expression and chromatin datasets are aligned to different assemblies. The source expression study used *Felis_catus_9.0*, whereas the chromatin study declares GCA_018350175.1 (*F.catus_Fca126_mat1.0*, provisionally felCat10) as its genome build. Both assemblies use the same chromosome names (A1, A2, B1 and so on), so peak coordinates translate onto a *Felis_catus_9.0* annotation without any name mismatch and without warning: in our initial attempt, 99.96% of peaks mapped successfully. That harmonisation is nevertheless incorrect, and we detected it only through a permutation control (Section 2.4), which showed promoter H3K4me3 enrichment of 1.55-fold, close to the value expected by chance. We report this because the same two datasets are likely to be combined by other groups and the failure is silent.

Because the expression matrices are indexed by gene symbol rather than by coordinate, the assembly used for expression quantification does not constrain the regulatory

annotation. We therefore annotated all genomic features on F.catus_Fca126_mat1.0, the assembly of the chromatin data, and joined the two layers by gene symbol. Peak chromosome names were translated to RefSeq accessions in two steps: the `(chr)` prefix was removed, and the resulting name was matched against the Sequence-Name, Assigned-Molecule and UCSC-style-name columns of the assembly report. Peaks on sequences absent from the chromosome-level annotation were discarded. narrowPeak intervals are zero-based and half-open, and were converted to one-based inclusive coordinates by adding one to the start. Because no blacklist exists for this species, we additionally discarded peaks in the upper 0.5% of the width distribution of each mark, where mapping artefacts concentrate. Retention after all filters was 99.46% for H3K4me3, 99.48% for H3K27ac and 99.46% for CTCF. Gene symbols in the expression matrices matched the RefSeq annotation for 99.87% of cortex genes and 99.88% of medulla genes.

2.3 Regulatory annotation

Promoters were defined as the interval from 2,000 bp upstream to 500 bp downstream of the annotated transcription start site, oriented by strand. Distal regulatory space was delimited by CTCF occupancy rather than by a fixed window: for each gene, the domain extended from the midpoint of the nearest CTCF peak upstream of the transcription start site to the midpoint of the nearest CTCF peak downstream, capped at 250 kb on each side to avoid unbounded domains in CTCF-poor regions. This follows the approach used by the FAANG consortium for domain prediction in domestic animal genomes. Resulting domains had a median width of 39.4 kb (interquartile range 12.8–102.3 kb; mean 79.2 kb), with a minority of genes reaching the 250 kb cap.

Each gene was assigned to one of four mutually exclusive classes in a fixed hierarchy: active promoter when both H3K4me3 and H3K27ac peaks overlapped the promoter window; primed promoter when H3K4me3 overlapped the promoter without H3K27ac; distal enhancer only when an H3K27ac peak fell within the CTCF-delimited domain outside any promoter window; and no active mark otherwise. Peak counts per mark within promoter and domain were retained as separate columns so that classes can be redefined without reprocessing.

2.4 Validation of the annotation

Because coordinate discordance between datasets is silent, we validated the annotation by permutation before any downstream analysis. For each mark, peaks were shuffled within chromosome, preserving peak number and width distribution, and the number of genes with at least one promoter-overlapping peak was recomputed; 100 permutations were used. Observed values were 17,419 genes for H3K4me3 (49.2% of annotated genes; 8.59-fold over the permuted mean), 17,256 for H3K27ac (3.92-fold) and 14,358 for CTCF (5.09-fold). The analysis pipeline aborts if H3K4me3 promoter enrichment falls below fivefold. The lower observed fold for H3K4me3 relative to values typical of human ENCODE-grade data reflects dilution by the large fraction of predicted and non-coding genes in the RefSeq annotation with imprecisely defined transcription start sites.

2.5 Differential expression

Because the deposited matrices are log2 counts per million rather than integer counts, negative-binomial frameworks are not applicable. Differential expression was assessed with limma, fitting a linear model per gene and applying empirical Bayes moderation with an intensity-dependent trend and robust estimation of the hyperparameters. The design included disease group without intercept and sex as a covariate; sex was retained because the medulla set is unbalanced (10 female, 8 male) even though the source study reported no sex effect. Age was not included in the primary model, for comparability

with the source study and because it is balanced across disease groups (Kruskal-Wallis $P = 0.87$ in the cortex and 0.86 in the medulla), but its influence was assessed by refitting each contrast with age as an additional covariate. In the cortex the two models agree closely, sharing 4,950 of the genes called by either at a false discovery rate below 0.10 (Jaccard index 0.94) with a Spearman correlation of 0.999 between fold changes. In the medulla the effect estimates are equally stable (correlation 0.998) but the age-adjusted model detects 3,578 genes against 1,908, retaining all genes called by the unadjusted model. Age therefore accounts for an appreciable share of residual variance in the medulla without altering the direction or magnitude of disease effects, and medullary counts should be read as conservative. Three contrasts were evaluated in each compartment: CKD 1/2 versus control, CKD 3/4 versus control, and CKD 3/4 versus CKD 1/2. Following the threshold used in the source study, genes were called differentially expressed at a Benjamini-Hochberg false discovery rate below 0.10.

At that threshold the cortex yielded 3 genes for the early contrast, 5,087 for the late contrast and none between stages; the medulla yielded 1,908, 2,821 and 8 respectively. These counts are concordant with the source study, which reported 6 and 4,616 for the cortical contrasts and approximately 2,000 and 2,484 for the medullary contrasts using edgeR with a quasi-likelihood negative binomial model. The between-stage contrast was near-empty in both compartments and was not carried forward; this decision was made after inspecting the result and is reported as such.

2.6 Coexpression module detection and stability

Coexpression structure was sought separately among induced and repressed genes, because correlation between a repressed and an induced gene is negative for reasons of direction rather than coregulation. Gene-gene distance was defined as one minus the Spearman correlation across samples of the relevant compartment, hierarchical clustering used average linkage, and modules were cut with the dynamic tree cut algorithm (deepSplit 2, minimum module size 30 genes).

Module reproducibility was assessed by subsampling animals rather than genes. In each of 200 iterations, 80% of samples were drawn without replacement, the clustering was recomputed, and each reference module was scored by its maximum Jaccard index against any module of the resampled solution. Modules with a mean Jaccard index below 0.6 were discarded. Module eigengenes were defined as the first principal component of the scaled expression of member genes, with sign fixed so that the eigengene increases when member genes increase, and their association with disease group was tested by Kruskal-Wallis with Benjamini-Hochberg correction.

In the cortex, using genes with a false discovery rate below 0.05 and absolute log₂ fold change above 1 (1,798 genes: 1,221 induced, 577 repressed), 13 modules were detected and 2 were retained: a 396-gene repressed module (mean Jaccard 0.731; first component explaining 78.4% of variance) and a 78-gene repressed module (0.631; 60.4%). No induced module reached the threshold. In the medulla, using the 1,908 genes of the early contrast, 21 modules were detected and none was retained.

We tested whether the asymmetry between directions was explained by measurement precision. Repressed genes were more abundant than induced genes (median average expression 4.61 versus 3.00; Wilcoxon $P < 10^{-15}$), but module stability was uncorrelated with module-level abundance (Spearman $\rho = -0.06$, $P = 0.85$), and a low-abundance repressed module was stable while a high-abundance induced module was not.

2.7 Cell-composition assessment

Composition was estimated with compartment-appropriate marker sets scored as the mean per-gene z-score across samples. For the cortex, the epithelial set comprised proximal tubule markers (LRP2, CUBN, SLC34A1, SLC5A2, SLC5A12, MIOX, GGT1, SLC22A6, SLC22A8, SLC3A1, SLC7A9, HNF4A). For the medulla, it comprised thick ascending limb, collecting duct, intercalated cell and thin limb markers (UMOD, SLC12A1, CLDN16, CLDN10, KCNJ1, ESRRB, AQP2, AQP3, SCNN1G, SCNN1B, GATA3, ELF5, ATP6V1B1, ATP6V0D2, SLC4A1, FOXI1, AQP1). Immune (PTPRC, CD68, CSF1R, ITGAM, AIF1, LYZ, C1QA, C1QB, TYROBP, FCER1G) and fibroblast (COL1A1, COL1A2, COL3A1, ACTA2, PDGFRB, FN1, LUM, DCN) sets were common to both compartments. Marker recovery is reported in the supplement.

Because the cohort spans 2 to 20 years of age, we also tested whether composition scores vary with age. None did, either marginally (Spearman rho = 0.04 for the tubular score, -0.19 for immune and -0.08 for fibroblast) or conditional on disease group (P = 0.29, 0.56 and 0.99). With 21 animals these tests have limited power, but the point estimates are close to zero.

The eigengene of the larger stable cortical module correlated with the proximal tubule score at rho = 0.891 ($P = 2.3 \times 10^{-6}$), with the immune score at -0.809 and with the fibroblast score at -0.869. Disease group explained 29.0% of the variance in the cortical tubule score and 18.5% of the variance in the medullary epithelial score (P = 0.215), so composition and stage are not collinear at the level of animals.

We considered adjusting differential expression for the marker-derived composition score and rejected the approach. The score is computed from the same expression matrix as the outcome and correlates with the dominant module eigengene at 0.891; including it as a covariate removes the principal axis of disease-related variation rather than a confounder, and the resulting loss of 97.5% of differentially expressed genes is an arithmetic consequence of that endogeneity rather than evidence about composition. No composition-adjusted differential expression is reported.

2.8 Motif enrichment

Position weight matrices were obtained from the JASPAR2020 CORE vertebrate collection via TFBSTools and restricted to matrices whose name corresponded to at least one gene present in the expression matrix of the compartment under analysis, leaving 401 of 633 matrices. Regulatory regions were defined as 300 bp windows centred on the peak summit, taken from the tenth column of the narrowPeak files; fixing the window width prevents differences in peak width between marks from being read as differences in motif content. Promoter foreground regions were H3K4me3 summits within the promoter window of genes in the query set; enhancer foreground regions were H3K27ac summits within the CTCF-delimited domain of those genes and outside any promoter window. Background regions were the corresponding regions of expressed protein-coding genes not differentially expressed in the same contrast, which controls simultaneously for base composition, for regulatory activity and for expression in the tissue. Sequences containing ambiguous bases were removed, sequences were assigned unique coordinate-derived names, foreground was capped at 1,500 regions and background at three times the foreground or 4,500 regions, whichever was smaller.

Enrichment was computed with monaLisa v1.18.0 using binned motif enrichment with the background bin as reference, Fisher testing and Benjamini-Hochberg correction within each analysis. GC bins were extended to span the full range from 0 to 1, because the default upper bound of 0.8 excludes CpG-island promoters; with the default breaks, 98 of 1,929 cortical repressed promoter regions fell outside the binned range and were

silently discarded. Each analysis was repeated after removing immune and fibroblast marker genes from the query set as a sensitivity analysis; because foreground regions are subsampled, the two runs draw different subsets and the comparison should be read as approximate.

Across twelve analyses covering two compartments, two directions and two region types, enrichment was largely null. Motif hits reaching a corrected P below 0.05 were confined to promoters and had log₂ enrichment values below 1.1, well below the range typical of driver motifs. Directional coherence between motif enrichment and the expression change of the corresponding factor was required for any candidate to be carried forward.

2.9 Characterisation of ERR-motif genes

Repressed cortical genes were partitioned by the presence of an ESRRB, ESRRB or ESRRG motif in the H3K4me₃-marked portion of their promoter, identified with monaLisa motif scanning at a minimum log-odds score of 10; 136 of 2,158 repressed genes carried such a site. Functional enrichment of the ERR-positive subset was tested with g:Profiler using the g:SCS correction and human orthologs, with all repressed genes as the custom background, so that the comparison is internal to the repressed programme rather than against the transcriptome. CTCF and SP1 motifs were processed identically as negative controls.

Because promoters of mitochondrial genes carry ERR elements constitutively, we tested whether the association is specific to the disease state. The odds of mitochondrial annotation (GO:0005739, including descendant terms, retrieved from org.Hs.eg.db) given an ERR site were 2.34 (95% CI 1.60–3.41) among repressed genes and 1.76 (1.34–2.30) among genes that were not repressed, with no evidence of interaction ($P = 0.213$); ERR-positive genes among the non-repressed were themselves enriched for aerobic respiration and the citrate cycle. We therefore inverted the outcome and modelled repression as a function of ERR-site presence by logistic regression, conditioning on mitochondrial annotation and on the logarithm of the number of scanned promoter regions per gene, the latter because genes with more promoter sequence have a higher probability of containing any motif.

2.10 Tissue breadth of regulatory elements

Each kidney peak was scored by the number of the seven profiled tissues in which an overlapping peak of the same mark was called, and classified as kidney-specific (one tissue), restricted (two to three), shared (four to five) or ubiquitous (six to seven). Genes were characterised by the mean tissue breadth of the H3K27ac peaks within their CTCF-delimited domain and outside promoter windows, by the fraction of those peaks that were kidney-specific, by the mean fold-enrichment of those peaks and by their number. Genes were classified as repressed, induced or unchanged from the cortical late contrast at a false discovery rate below 0.10.

Association with expression state was assessed by Kruskal-Wallis across the three states, by pairwise Wilcoxon tests against the unchanged set, and by logistic regression of expression state on the kidney-specific fraction adjusted for the logarithm of enhancer number. Because peak signal intensity correlates with apparent tissue breadth (Spearman $\rho = 0.424$, $P < 10^{-15}$), models were refitted with mean enhancer fold-enrichment as an additional covariate.

2.11 Reporting of analytical decisions

This study was not conducted under a prespecified analysis plan and is reported as exploratory. Several analytical decisions were made in response to intermediate results

and are disclosed here: the between-stage contrast was dropped after it returned almost no genes in either compartment; the primary compartment for the regulatory analyses was chosen after examining differential expression in both; coexpression modules were abandoned as an organising unit after they failed the stability criterion; and the false discovery threshold of 0.10 was adopted to match the source study rather than selected on the basis of our own results. The stability threshold of 0.6, the promoter and domain window parameters, the permutation validation and the confounder adjustments were fixed before the analyses they govern.

2.12 Software and availability

All analyses were performed in R [version]. Genomic intervals were handled with GenomicRanges and rtracklayer, sequence retrieval with Rsamtools and Biostrings, differential expression with limma, clustering with dynamicTreeCut, motif analysis with monaLisa v1.18.0, TFBSTools and JASPAR2020, functional enrichment with gprofiler2 against the human namespace, gene ontology retrieval with org.Hs.eg.db, and figures with ggplot2 and patchwork. Analysis code is organised as sequentially numbered modules and is available at [repository URL] with a persistent identifier at [DOI].